

Total no. of Pages:

Roll no.

Degree: B.Tech. ECAM Semester: Third
MID-SEMESTER EXAMINATION, June, 2026

Course Title: Digital Circuits and Systems
Course Code: EAEPC305

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Explain in detail the theory and operation of TTL NAND Gate.	2	CO1
1b	Express the Boolean function $F = A + B'C$ as a (i) sum of Minterms (ii) product of Maxterms.	2	CO1
2a	Implement Boolean function using 4:1 MUX: $F(A, B, C) = \Sigma m(1, 2, 6, 7)$	2	CO2
2b	Implement: $F = AB + AC + BC$ using decoder.	2	CO2
3a	Add 523 and 679 using BCD addition.	2	CO1
3b	Design and draw the logic diagram of Full adder and full subtractor.	2	CO2
4a	Differentiate between PROM, PLA and PAL.	2	CO2
4b	Implement a 8:1 multiplexer using 4:1 multiplexer.	2	CO2
5a	Design a combinational circuit whose input is a four-bit number and whose output is the 2's complement of the input number.	2	CO2
5b	Convert $(7A3)_{16}$ into binary and octal.	2	CO1